

lme4 and Beyond

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Outline of Talk

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- Data & Analyses
- Results

Regression Models

Linear Mixed-Effects Regression Model

A **linear mixed-effects regression** (LMER) model can be written as

$$\mathbf{y} = \underbrace{\mathbf{X}\boldsymbol{\beta}}_{\text{fixed}} + \underbrace{\mathbf{Z}\mathbf{u}}_{\text{random}} + \boldsymbol{\epsilon}$$

where

- $\mathbf{y} = (y_1, \dots, y_n)' \in \mathbb{R}^n$ is the response vector
- $\mathbf{X} = \{x_{ij}\}_{n \times p} \in \mathcal{X}$ is the known fixed-effects design matrix
- $\boldsymbol{\beta} \in \mathbb{R}^p$ are the unknown fixed-effects coefficients
- $\mathbf{Z} = \{z_{ij}\}_{n \times q} \in \mathcal{Z}$ is the known random-effects design matrix
- $\mathbf{u} \sim \mathbf{N}(\mathbf{0}, \mathbf{G})$ are the unknown Gaussian random effects
 - $\mathbf{G} = \mathbf{G}(\boldsymbol{\phi}_G)$ where $\boldsymbol{\phi}_G$ are the unique parameters
- $\boldsymbol{\epsilon} \sim \mathbf{N}(\mathbf{0}, \mathbf{R})$ is unknown Gaussian error
 - $\mathbf{R} = \mathbf{R}(\boldsymbol{\phi}_R)$ where $\boldsymbol{\phi}_R$ are the unique parameters
- $\mathbf{u}$ and $\boldsymbol{\epsilon}$ are independent, i.e., $E(\mathbf{u}\boldsymbol{\epsilon}') = \mathbf{0}$ ($q \times n$ matrix of zeros)

Joint Log-Likelihood Function

The joint density has the form $f(\mathbf{y}, \mathbf{u}) = f(\mathbf{y}|\mathbf{u})f(\mathbf{u})$.

- $f(\mathbf{y}|\mathbf{u}) \equiv \mathbf{N}(\mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u}, \mathbf{R})$
- $f(\mathbf{u}) \equiv \mathbf{N}(\mathbf{0}, \mathbf{G})$
- $f(\mathbf{y}, \mathbf{u}) \equiv \mathbf{N}\left(\begin{pmatrix} \mathbf{X}\boldsymbol{\beta} \\ \mathbf{0} \end{pmatrix}, \begin{pmatrix} \mathbf{Z}\mathbf{G}\mathbf{Z}' + \mathbf{R} & \mathbf{Z}\mathbf{G} \\ \mathbf{G}\mathbf{Z}' & \mathbf{G} \end{pmatrix}\right)$

Dropping constants, the joint log-likelihood has the form

$$l_J(\boldsymbol{\beta}, \phi_R, \phi_G | \mathbf{y}, \mathbf{u}) = -\frac{1}{2}\mathbf{r}'\mathbf{R}^{-1}\mathbf{r} - \frac{1}{2}\mathbf{u}'\mathbf{G}^{-1}\mathbf{u}$$

where $\mathbf{r} = \mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u}$.

Henderson's (1950) Mixed-Model Equations

The coefficients $(\hat{\beta}, \hat{\mathbf{u}})$ that maximize $l_J(\beta, \mathbf{u} | \mathbf{y}, \phi_R, \phi_G)$ satisfy

$$\begin{pmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{X}'\mathbf{R}^{-1}\mathbf{Z} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z} + \mathbf{G}^{-1} \end{pmatrix} \begin{pmatrix} \hat{\beta} \\ \hat{\mathbf{u}} \end{pmatrix} = \begin{pmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{y} \end{pmatrix}$$

where

- $\hat{\beta}$ is the empirical best linear unbiased estimator (BLUE) of β
- $\hat{\mathbf{u}}$ is the empirical best linear unbiased predictor (BLUP) of $\mathbf{u}$

Defining $\mathbf{V} = \mathbf{ZGZ}' + \mathbf{R}$, the optimal coefficients have the form

$$\begin{aligned} \hat{\beta} &= (\mathbf{X}'\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}'\mathbf{V}^{-1}\mathbf{y} \\ \hat{\mathbf{u}} &= \mathbf{GZ}'\mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\hat{\beta}) \end{aligned}$$

where $\hat{\mathbf{u}} = E(\mathbf{u} | \mathbf{y})$ is the empirical expected value of $\mathbf{u}$ given $\mathbf{y}$.

Full and Restricted Log-Likelihood Functions

$$(\mathbf{y}|\boldsymbol{\beta}, \boldsymbol{\phi}_R, \boldsymbol{\phi}_G) \sim \mathbf{N}(\mathbf{X}\boldsymbol{\beta}, \mathbf{V}) \text{ where } \mathbf{V} = \mathbf{ZGZ}' + \mathbf{R}$$

Dropping constant terms, the full and restricted (REML) log-likelihoods are

$$l_F(\boldsymbol{\beta}, \boldsymbol{\phi}_R, \boldsymbol{\phi}_G|\mathbf{y}) = -\frac{1}{2} \log(|\mathbf{V}|) - \frac{1}{2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'\mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

$$l_R(\boldsymbol{\beta}, \boldsymbol{\phi}_R, \boldsymbol{\phi}_G|\mathbf{y}) = l_F(\boldsymbol{\beta}, \boldsymbol{\phi}_R, \boldsymbol{\phi}_G|\mathbf{y}) - \frac{1}{2} \log(|\mathbf{X}'\mathbf{V}^{-1}\mathbf{X}|)$$

Parameters $\{\boldsymbol{\beta}, \boldsymbol{\phi}_R, \boldsymbol{\phi}_G\}$ estimated using iterative algorithm (FS, NR, EM).

Nonparametric Mixed-Effects Regression Model

A nonparametric mixed-effects regression (NPMER) model can be written as

$$\mathbf{y} = \underbrace{\mathbf{X}\boldsymbol{\beta}}_{\text{fixed}} + \underbrace{\boldsymbol{\eta}}_{\text{???}} + \underbrace{\mathbf{Z}\mathbf{u}}_{\text{random}} + \boldsymbol{\epsilon}$$

where

- $\mathbf{y} = (y_1, \dots, y_n)' \in \mathbb{R}^n$ is the response vector
- $\mathbf{X} = \{x_{ij}\}_{n \times p} \in \mathcal{X}$ is the known fixed-effects design matrix
- $\boldsymbol{\beta} \in \mathbb{R}^p$ are the unknown fixed-effects coefficients
- $\boldsymbol{\eta} = \{\eta(\mathbf{x}_i)\}_{n \times 1}$ with $\eta \in \mathcal{H}$ denoting an unknown function
- $\mathbf{Z} = \{z_{ij}\}_{n \times q} \in \mathcal{Z}$ is the known random-effects design matrix
- $\mathbf{u} \sim \mathbf{N}(\mathbf{0}, \mathbf{G})$ are the unknown Gaussian random effects [$\mathbf{G} = \mathbf{G}(\phi_G)$]
- $\boldsymbol{\epsilon} \sim \mathbf{N}(\mathbf{0}, \mathbf{R})$ is unknown Gaussian error [$\mathbf{R} = \mathbf{R}(\phi_R)$]
- $\mathbf{u}$ and $\boldsymbol{\epsilon}$ are independent, i.e., $E(\mathbf{u}\boldsymbol{\epsilon}') = \mathbf{0}$ ($q \times n$ matrix of zeros)

Representing the Unknown Function

Suppose that $\mathcal{H}$ is a reproducing kernel Hilbert space

- $\langle \cdot, \cdot \rangle$ denotes the inner product
- $\rho(\cdot, \cdot)$ denotes the reproducing kernel (Aronszajn, 1950)
- $\langle \rho(\mathbf{x}, \tilde{\mathbf{x}}), \eta(\tilde{\mathbf{x}}) \rangle = \eta(\mathbf{x})$ for all $\mathbf{x}, \tilde{\mathbf{x}} \in \mathcal{X}$ and $\eta \in \mathcal{H}$

$\eta(\mathbf{x}) = \sum_{j=1}^r c_j \rho(\mathbf{x}, \tilde{\mathbf{x}}_j)$ where $\{\tilde{\mathbf{x}}_j\}_{j=1}^r \subset \{\mathbf{x}_i\}_{i=1}^n$ are the “knots”

- KWRT: use all $\mathbf{x}_i$ as knots (Kimeldorf & Wahba, 1971)
- Strategic knots perform well (Helwig & Ma, 2015; Ma et al., 2015).

$\eta = \mathbf{J}\mathbf{c}$ and $\langle \eta, \eta \rangle = \mathbf{c}'\mathbf{Q}\mathbf{c}$ where $\mathbf{J} = \{\rho(\mathbf{x}_i, \tilde{\mathbf{x}}_j)\}_{n \times r}$, $\mathbf{Q} = \{\rho(\tilde{\mathbf{x}}_j, \tilde{\mathbf{x}}_k)\}_{r \times r}$.

- $\mathbf{J} = \mathbf{J}(\phi_Q)$ and $\mathbf{Q} = \mathbf{Q}(\phi_Q)$ where ϕ_Q are the unique parameters

Joint Log-Likelihood Function

The joint density has the form $f(\mathbf{y}, \mathbf{c}, \mathbf{u}) = f(\mathbf{y}|\mathbf{c}, \mathbf{u})f(\mathbf{c})f(\mathbf{u})$.

- $f(\mathbf{y}|\mathbf{c}, \mathbf{u}) \equiv N(\mathbf{X}\boldsymbol{\beta} + \mathbf{J}\mathbf{c} + \mathbf{Z}\mathbf{u}, \mathbf{R})$
- $f(\mathbf{c}) \equiv N(\mathbf{0}, \mathbf{Q}^\dagger)$
- $f(\mathbf{u}) \equiv N(\mathbf{0}, \mathbf{G})$
- $f(\mathbf{y}, \mathbf{c}, \mathbf{u}) \equiv N \left(\begin{pmatrix} \mathbf{X}\boldsymbol{\beta} \\ \mathbf{0} \\ \mathbf{0} \end{pmatrix}, \begin{pmatrix} \mathbf{J}\mathbf{Q}^\dagger\mathbf{J}' + \mathbf{Z}\mathbf{G}\mathbf{Z}' + \mathbf{R} & \mathbf{J}\mathbf{Q}^\dagger & \mathbf{Z}\mathbf{G} \\ \mathbf{Q}^\dagger\mathbf{J}' & \mathbf{Q}^\dagger & \mathbf{0} \\ \mathbf{G}\mathbf{Z}' & \mathbf{0} & \mathbf{G} \end{pmatrix} \right)$

Dropping constants, the joint log-likelihood has the form

$$l_J(\boldsymbol{\beta}, \phi_R, \phi_G, \phi_Q | \mathbf{y}, \mathbf{c}, \mathbf{u}) = -\frac{1}{2}\mathbf{r}'\mathbf{R}^{-1}\mathbf{r} - \frac{1}{2}\mathbf{c}'\mathbf{Q}\mathbf{c} - \frac{1}{2}\mathbf{u}'\mathbf{G}^{-1}\mathbf{u}$$

where $\mathbf{r} = \mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{J}\mathbf{c} - \mathbf{Z}\mathbf{u}$.

Revisiting Henderson's (1950) Mixed-Model Equations

The coefficients $(\hat{\beta}, \hat{\mathbf{c}}, \hat{\mathbf{u}})$ that maximize $l_J(\beta, \mathbf{c}, \mathbf{u} | \mathbf{y}, \phi_R, \phi_G, \phi_Q)$ satisfy

$$\begin{pmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{X}'\mathbf{R}^{-1}\mathbf{J} & \mathbf{X}'\mathbf{R}^{-1}\mathbf{Z} \\ \mathbf{J}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{J}'\mathbf{R}^{-1}\mathbf{J} + \mathbf{Q} & \mathbf{J}'\mathbf{R}^{-1}\mathbf{Z} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{X} & \mathbf{Z}'\mathbf{R}^{-1}\mathbf{J} & \mathbf{Z}'\mathbf{R}^{-1}\mathbf{Z} + \mathbf{G}^{-1} \end{pmatrix} \begin{pmatrix} \hat{\beta} \\ \hat{\mathbf{c}} \\ \hat{\mathbf{u}} \end{pmatrix} = \begin{pmatrix} \mathbf{X}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{J}'\mathbf{R}^{-1}\mathbf{y} \\ \mathbf{Z}'\mathbf{R}^{-1}\mathbf{y} \end{pmatrix}$$

where

- $\hat{\beta}$ is the empirical best linear unbiased estimator (BLUE) of β
- $\hat{\mathbf{c}}$ is the empirical best linear unbiased predictor (BLUP) of $\mathbf{c}$
- $\hat{\mathbf{u}}$ is the empirical best linear unbiased predictor (BLUP) of $\mathbf{u}$

Defining $\mathbf{V} = \mathbf{ZGZ}' + \mathbf{R}$, the optimal coefficients have the form

$$\begin{pmatrix} \hat{\beta} \\ \hat{\mathbf{c}} \end{pmatrix} = \begin{pmatrix} \mathbf{X}'\mathbf{V}^{-1}\mathbf{X} & \mathbf{X}'\mathbf{V}^{-1}\mathbf{J} \\ \mathbf{J}'\mathbf{V}^{-1}\mathbf{X} & \mathbf{J}'\mathbf{V}^{-1}\mathbf{J} + \mathbf{Q} \end{pmatrix}^{\dagger} \begin{pmatrix} \mathbf{X}'\mathbf{V}^{-1}\mathbf{y} \\ \mathbf{J}'\mathbf{V}^{-1}\mathbf{y} \end{pmatrix}$$

$$\hat{\mathbf{u}} = \mathbf{GZ}'\mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\hat{\beta} - \mathbf{J}\hat{\mathbf{c}})$$

Penalized Likelihood Estimation

$$(\mathbf{y}|\boldsymbol{\beta}, \mathbf{c}, \phi_R, \phi_G) \sim N(\mathbf{X}\boldsymbol{\beta} + \mathbf{J}\mathbf{c}, \mathbf{V}) \text{ where } \mathbf{V} = \mathbf{Z}\mathbf{G}\mathbf{Z}' + \mathbf{R}$$

Dropping constant terms, the penalized log-likelihoods are

$$l_F(\boldsymbol{\beta}, \mathbf{c}, \phi_R, \phi_G, \phi_Q|\mathbf{y}) = -\frac{1}{2} \log(|\mathbf{V}|) - \frac{1}{2} \|\mathbf{V}^{-1/2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{J}\mathbf{c})\|^2 - \frac{1}{2} \mathbf{c}'\mathbf{Q}\mathbf{c}$$

$$l_R(\boldsymbol{\beta}, \mathbf{c}, \phi_R, \phi_G, \phi_Q|\mathbf{y}) = l_F(\boldsymbol{\beta}, \mathbf{c}, \phi_R, \phi_G, \phi_Q|\mathbf{y}) - \frac{1}{2} \log(|\tilde{\mathbf{X}}'\mathbf{V}^{-1}\tilde{\mathbf{X}}|) - \frac{1}{2} \mathbf{c}'\mathbf{Q}\mathbf{c}$$

where $\tilde{\mathbf{X}} = [\mathbf{X}, \mathbf{J}]$ is the combined fixed-effect design matrix.

Given $\{\phi_R, \phi_G, \phi_Q\}$, the penalized likelihood estimates of $\boldsymbol{\beta}$ and $\mathbf{c}$ are identical to those solving the (extended) mixed-model equations.

Variance and/or Smoothing Parameter Estimation

How should we estimate $\phi = \{\phi_R, \phi_G, \phi_Q\}$?

Three general approaches:

- ① Treat all parameters in ϕ as variance parameters
 - $\mathbf{y} \sim N(\mathbf{X}\beta, \mathbf{JQ}^\dagger\mathbf{J}' + \mathbf{ZGZ}' + \mathbf{R})$
 - REML estimation of all parameters in ϕ
 - Wang (1998a) and Wang (1998b)
- ② Treat $\{\phi_Q, \phi_G\}$ as smoothing parameters and ϕ_R as variance parameters
 - $(\mathbf{y}|\mathbf{c}, \mathbf{u}) \sim N(\mathbf{X}\beta + \mathbf{Jc} + \mathbf{Zu}, \mathbf{R})$
 - Generalized cross-validation (GCV) estimation of $\{\phi_Q, \phi_G\}$
 - Gu and Ma (2005)
- ③ Treat ϕ_Q as smoothing parameters and $\{\phi_R, \phi_G\}$ as variance parameters
 - $(\mathbf{y}|\mathbf{c}) \sim N(\mathbf{X}\beta + \mathbf{Jc}, \mathbf{ZGZ}' + \mathbf{R})$
 - GCV of ϕ_Q and REML of $\{\phi_R, \phi_G\}$
 - Helwig (2016)

NPMER in Practice

Overview of Regression Functions in R

Different Types of Regression

- Linear: coefficients enter the model linearly
 - $\hat{Y} = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$
 - `lm` and `lmer` in *lme4* (Bates et al., 2015)
- Nonlinear: coefficients enter the model nonlinearly
 - $\hat{Y} = \sin(\beta_0 + 2\pi\beta_1 X)$
 - `nlm` and `nlmer` in *lme4* (Bates et al., 2015)
- Nonparametric: unspecified form for regression function
 - $\hat{Y} = \eta(X)$
 - `gam` in *mgcv* (Wood, 2016)
 - `ssanova` in *gss* (Gu, 2014)
 - `bigssp` in *bigsplines* (Helwig, 2017)

Steps for Fitting NPMER Model

- 1 Choose type (fixed vs random) for each predictor
- 2 Specify covariance structure for random effects
- 3 Specify the fixed-effect formula
- 4 Choose spline type for each nonparametric predictor
- 5 Sample representative spline knots

Note that steps 1–3 are done in LMER too (steps 4–5 are unique to NPMER).

3) Specifying the Nonparametric (“Fixed-Effects”) Formula

With $p > 1$ predictors, we need to specify the form of the function.

For example, with $p = 2$ predictors we have two possible models

additive : $\eta(x_1, x_2) = \eta_0 + \eta_1(x_1) + \eta_2(x_2)$

interactive : $\eta(x_1, x_2) = \eta_0 + \eta_1(x_1) + \eta_2(x_2) + \eta_{12}(x_1, x_2)$

where

- η_0 is an analogue of β_0 (unknown regression intercept)
- η_1 is the unknown main effect function for the first predictor
- η_2 is the unknown main effect function for the second predictor
- η_{12} is the unknown interaction effect function

4) Penalty Types for Nonparametric Predictors

Domain	R class	Spline Type	Penalty
$x \in \{1, \dots, g\}$	factor	Nominal	$\sum_{x=1}^g [\eta(x) - \bar{\eta}]^2$
$x \in \{1 < \dots < g\}$	ordered	Ordinal	$\sum_{x=2}^g [\eta(x) - \eta(x-1)]^2$
$x \in [a, b]$	numeric [†]	Linear [‡]	$\int [\eta^{(1)}(x)]^2 \mathbf{d}x$
$x \in [a, b]$	numeric [†]	Cubic [‡]	$\int [\eta^{(2)}(x)]^2 \mathbf{d}x$
$\mathbf{x} \in \mathbb{R}^2$	matrix	Thin-Plate	$J_2(\eta)$
$\mathbf{x} \in \mathbb{R}^3$	matrix	Thin-Plate	$J_3(\eta)$

[†] Can be applied to integer class variables too.

[‡] Can be constrained to be periodic, i.e., such that $\eta(a) = \eta(b)$.

For $\mathbf{x} = (x_1, x_2)$, the cubic thin-plate spline penalty has the form

$$J_2(\eta) = \int \left(\frac{\partial^2 \eta(\mathbf{x})}{\partial x_1^2} \right)^2 + \left(\frac{\partial^2 \eta(\mathbf{x})}{\partial x_2^2} \right)^2 + 2 \left(\frac{\partial^2 \eta(\mathbf{x})}{\partial x_1 \partial x_2} \right)^2 \mathbf{d}x$$

5) Sampling Representative Spline Knots

The unknown function η depends on chosen knots

- $\mathbf{J} = \{\rho(\mathbf{x}_i, \tilde{\mathbf{x}}_j)\}_{n \times r}$ with ρ denoting a **reproducing kernel** function
- $\{\tilde{\mathbf{x}}_j\}_{j=1}^r \subset \{\mathbf{x}_i\}_{i=1}^n$ are the user-selected **spline knots**

Optimal solution uses all $\mathbf{x}_i$ as knots, which is infeasible for large samples.

Use random (Kim & Gu, 2004) or bin-sampled (Helwig & Ma, 2015) knots.

- Need to include enough knots for stable solution
- Be careful when sampling knots for a `factor`—include all levels!

Educational Achievement Example

Overview of Education Data from Paterson (1991)

Exam scores for $n = 3435$ students who attended 19 secondary schools that are cross classified with 148 primary schools (see Goldstein, 2011, Chap. 12).

We fit a NPMER model of the form

$$y_i = \eta(x_i) + u_{p_i1} + u_{s_i2} + \epsilon_i$$

for $i \in \{1, \dots, 3435\}$ where

- y_i is the secondary school leaving exam score
- x_i is a verbal reasoning exam score obtained in primary school
- $p_i \in \{1, \dots, 148\}$ is the primary school indicator
- $s_i \in \{1, \dots, 19\}$ is the secondary school indicator
- $u_{p1} \stackrel{\text{iid}}{\sim} \text{N}(0, \sigma^2 \tau_p)$, $u_{s2} \stackrel{\text{iid}}{\sim} \text{N}(0, \sigma^2 \tau_s)$, and $\epsilon_i \stackrel{\text{iid}}{\sim} \text{N}(0, \sigma^2)$

Look at the Data in R

```
> head(exam)
  VRQ Exam Primary Secondary pupil
1 111   10        1          9     1
2 100    3        1          9     2
3  86    2        1          9     3
4  94    3        1          9     4
5  70    2        1          9     5
6  83    2        1          9     6
> exam$Primary <- factor(exam$Primary)
> exam$Secondary <- factor(exam$Secondary)
> range(exam$VRQ)
[1]  70 140
```

Fit NPMER and LMER Models

```
# select 10 (approximately) equidistant knots
knot10 <- round(seq(70, 140, length.out=10))
knot10ID <- NULL
for(k in 1:10) {
  knot10ID <- c(knot10ID, which(exam$VRQ==knot10[k])[1])
}

# fit NPMER model
require(bigsplines)
npmod10 <- bigssp(Exam ~ VRQ, data = exam,
                  nknots = knot10ID, type = "cub",
                  random = ~ (1|Primary) + (1|Secondary))

# fit LMER model
require(lme4)
lmod <- lmer(Exam ~ VRQ + (1|Primary) + (1|Secondary),
            data = exam)
```

Estimated Functional Relationship

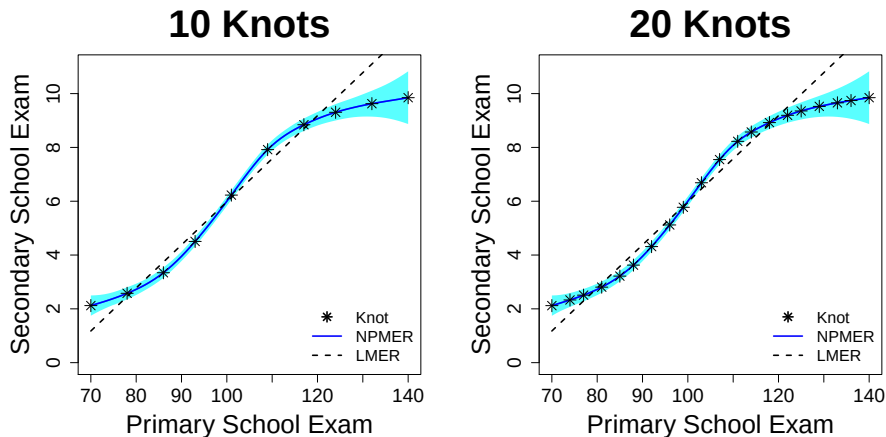


Figure 1: Predicted relationship between primary and secondary school exam.

Estimated Variance Components

Table 1: Variance parameter estimates for educational data.

Model	$\hat{\tau}_p$	$\hat{\tau}_s$	$\hat{\sigma}$
NPMER 10 knots	0.0672	0.0057	2.0171
NPMER 20 knots	0.0673	0.0053	2.0173
LMER	0.0646	0.0034	2.0627

Note. $\hat{\tau}_p$ and $\hat{\tau}_s$ are Primary and Secondary (respectively).

$\hat{\tau}_s < \hat{\tau}_p \implies$ more variation due to primary school effect

$\hat{\tau}_s < \hat{\tau}_p \ll 1 \implies$ more variation within the schools than between the schools

Electroencephalography (EEG) Example

EEG Data Source

EEG data from the UCI Machine Learning Repository (Lichman, 2013).

Collected by Henri Begleiter at the Neurodynamics Laboratory at the State University of New York Health Center at Brooklyn.

Event related potentials (ERPs) recorded from alcoholic and control subjects.

- Alcoholics were abstinent for a minimum of three weeks
- Part of a larger study examining EEG correlates of alcoholism

Visual Stimulus ERP Study

Experiment Details:

- Data from 120 subjects
 - 76 alcoholics
 - 44 controls
- 64-channel EEG cap
 - Ground: Nz
 - Reference: Cz
- Snodgrass & Vanderwart
 - Set of 260 pictures
- 10 trials sampled at 256 Hz for one second after stimulus

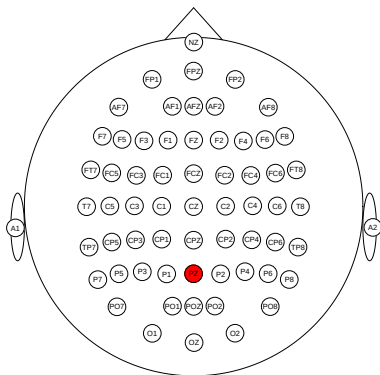


Figure 2: The 64-channel EEG cap.

Snodgrass & Vanderwart (1980) Stimuli

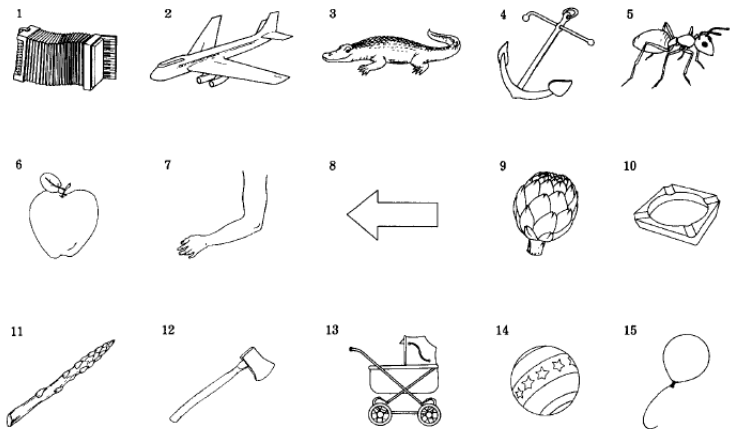


Figure 3: First 15 stimuli from Snodgrass & Vanderwart set.

NPMER Model for EEG Data

We fit a NPMER model of the form

$$y_i = \eta(t_i, g_i) + u_{s_i g_i} + \epsilon_i$$

where

- y_i is the recorded EEG voltage from electrode Pz
- $t_i \in \{1, \dots, 256\}$ is the time point at which y_i was recorded
- $g_i \in \{\text{alcoholic}, \text{control}\}$ is the group indicator
- $s_i \in \{1, \dots, 120\}$ is the subject indicator
- $u_{sg} \stackrel{\text{ind}}{\sim} \text{N}(0, \sigma^2 \tau_g)$ are the latent random intercepts
- $\epsilon_i \stackrel{\text{iid}}{\sim} \text{N}(0, \sigma^2)$ are latent error terms

Two-Way Smoothing Spline ANOVA Decomposition

The function η is assumed to have the form

$$\eta(t_i, g_i) = \eta_0 + \eta_T(t_i) + \eta_G(g_i) + \eta_{TG}(t_i, g_i)$$

where

- η_0 is an unknown constant (intercept) term
- η_T is the main effect of time
- η_G is the main effect of group
- η_{TG} is the time*group interaction effect

Using a **tensor product smoothing spline** with a cubic marginal (for time) and a nominal marginal (for group).

NPMER Model for EEG Data in R

```
### look at data
> head(eegdata)
  subject group condition trial channel time voltage
1         1     a     Slobj      0       PZ      0  -2.797
2         1     a     Slobj      0       PZ      1  -4.262
3         1     a     Slobj      0       PZ      2  -4.262
4         1     a     Slobj      0       PZ      3  -2.797
5         1     a     Slobj      0       PZ      4  -0.844
6         1     a     Slobj      0       PZ      5   0.132

### get 30 equidistant knots per group
> tvals <- round(seq(1,256,length.out=30))
> knotid <- c(which(eegdata$group=="a")[tvals],
+             which(eegdata$group=="c")[tvals])

### fit NPMER model
> eegmod <- bigssp(voltage ~ time * group, data = eegdata,
+                 type = list(time="cub", group="nom"),
+                 random = ~ group|subject, nknots = knotid)
```


Estimated Functional Relationship

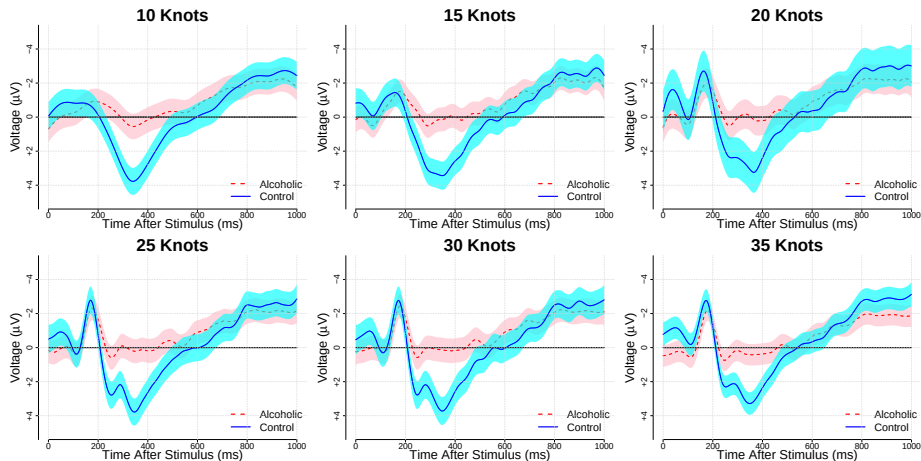


Figure 4: Predicted EEG for alcoholics and controls.

Estimated Variance Components

Table 2: Variance parameter estimates for EEG data.

Model	$\hat{\tau}_a$	$\hat{\tau}_c$	$\hat{\sigma}$
10 Knots	0.187	0.126	8.692
15 Knots	0.155	0.142	8.688
20 Knots	0.221	0.516	8.682
25 Knots	0.154	0.122	8.680
30 Knots	0.155	0.128	8.680
35 Knots	0.153	0.123	8.680

Note. $\hat{\tau}_a$ and $\hat{\tau}_c$ are Alcoholics and Controls (respectively).

$\hat{\tau}_a > \hat{\tau}_c \implies$ slightly more variation in alcoholics' data

$\hat{\tau}_c < \hat{\tau}_a < 1 \implies$ more variation within ERPs than between ERPs

Take Home Points

NPMER models offer a flexible framework for data analysis and prediction.

- Applicable to any LMER situation
- Requires fewer assumptions than LMER

Fitting NPMER models is (basically) as simple as fitting LMER models.

- Similar syntax for specifying the model
- Only new steps are spline type and knot selection

Shift in focus from thinking only about $\hat{\beta}$ to thinking about $\hat{\beta}$ and $\hat{\eta}$

- No longer only concerned with inference on coefficients
- Instead focus on prediction and visualization

References

- Aronszajn, N. (1950). Theory of reproducing kernels. *Transactions of the American Mathematical Society* 68, 337–404.
- Bates, D., M. Mächler, B. M. Bolker, and S. C. Walker (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software* 67, 1–48.
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